

AI6132 Generative AI For Visual Synthesis Course Project



Project 1: Technical Review

- Group based project
 - Each group has up to 4 members
 - You can form a group with fewer than 4 members.
 - The assessment criteria will be the same for any group size.
- Grouping:
 - please edit the online form below to create your group.
https://docs.google.com/spreadsheets/d/1a9JWn7yKkNpSGu1mHscrQmW6FtnU3o5ywBya_E78nBg/edit?usp=sharing
 - **Please enter your full name exactly as shown on your student card (important!)**
 - The first person for each group is the coordinator.

- Please complete the grouping by Mon Week 3
 - **Everyone should enter your name in the online form.**
 - If you do not join any existing group, please create a new group and enter your name. We will merge small groups if necessary.
 - If your group is not full (fewer than 4 members), additional members may be added when we finalize the groups.
 - The final grouping will be released in Week 3. You need to verify your grouping.

Submission

- Please submit 3 files using the following naming guidelines:
 - 1 “group_XX_report.pdf”
 - Your main report PDF, XX is your group ID.
 - 2. “group_XX_contribution.pdf”
 - A PDF describes the contribution for each member
 - 3. “group_XX_code.zip”
 - A ZIP package for the source code
 - no datasets, no models, no external links
- Where to submit
 - Please submit your zip file in NTULearn:
the course site -> Assignments -> submission link for project 1

Submission

- Submission requirement
 - One group only makes one submission.
 - You can submit multiple times (no limit).
 - You don't need to provide comments in your code.
 - You don't need to submit your **datasets or models**.
 - **Should not provide external links!**
 - cloud-hosted files are not accepted!
- Submission due date: 16 Nov 2026 (Mon, week 14)

Requirements

- A technical review of visual generation methods
- Identify a visual generation task.

Examples:

- New generative model paradigms
- Image/Video/3D/4D generation
- Image/Video/3D/4D Editing
- Controllable generation
- Efficient generation

- Use at least one dataset for evaluation
 - You can use a subset of a dataset
 - You can build your own dataset
 - Using more datasets is encouraged.
- Select methods for comparison and analysis
 - Select at least two methods
 - You can use any existing methods
 - You can propose your own methods.

- Evaluate and analyze the selected methods
 - Analysis of the hyper-parameter settings
 - Analysis of the key components in a method.
 - Comparison between the selected methods.
 - Discussion on the strengths and limitations.
 - Visualization and discussion of the results
 - Discussion on future extension.
 - Cannot simply copy results from other materials.
- Technical report and the code
 - At least 4 pages.
 - Format: introduction/methods/experiments and analysis/conclusion

Rubrics

A good report should meet the following:

- Meet the requirements well.
- Well organized and structured
- Clear explanation of the tasks and the methods.
- Comprehensive and in-depth comparisons and analysis.
- Present clear and insightful discussions
- Good visualizations for analysis.
- Target challenging datasets/applications/methods.
- Discuss or implement possible solutions to improve the methods

About teamwork

Responsibilities for every team member:

1. Ensure all members work together and everyone is involved and contributes to the project.
2. Sufficient coordination and communication in the team.
 - If a team member fails to contribute due to insufficient communication and coordination, the entire team will face penalties.
3. Report an issue
 - If a team member refuses to contribute, please email the instructors at least 6 weeks before the submission deadline.
 - If you cannot report the issue in time, it indicates that the entire team lacks communication and coordination. We will not handle any issues if they are not reported in time.

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